

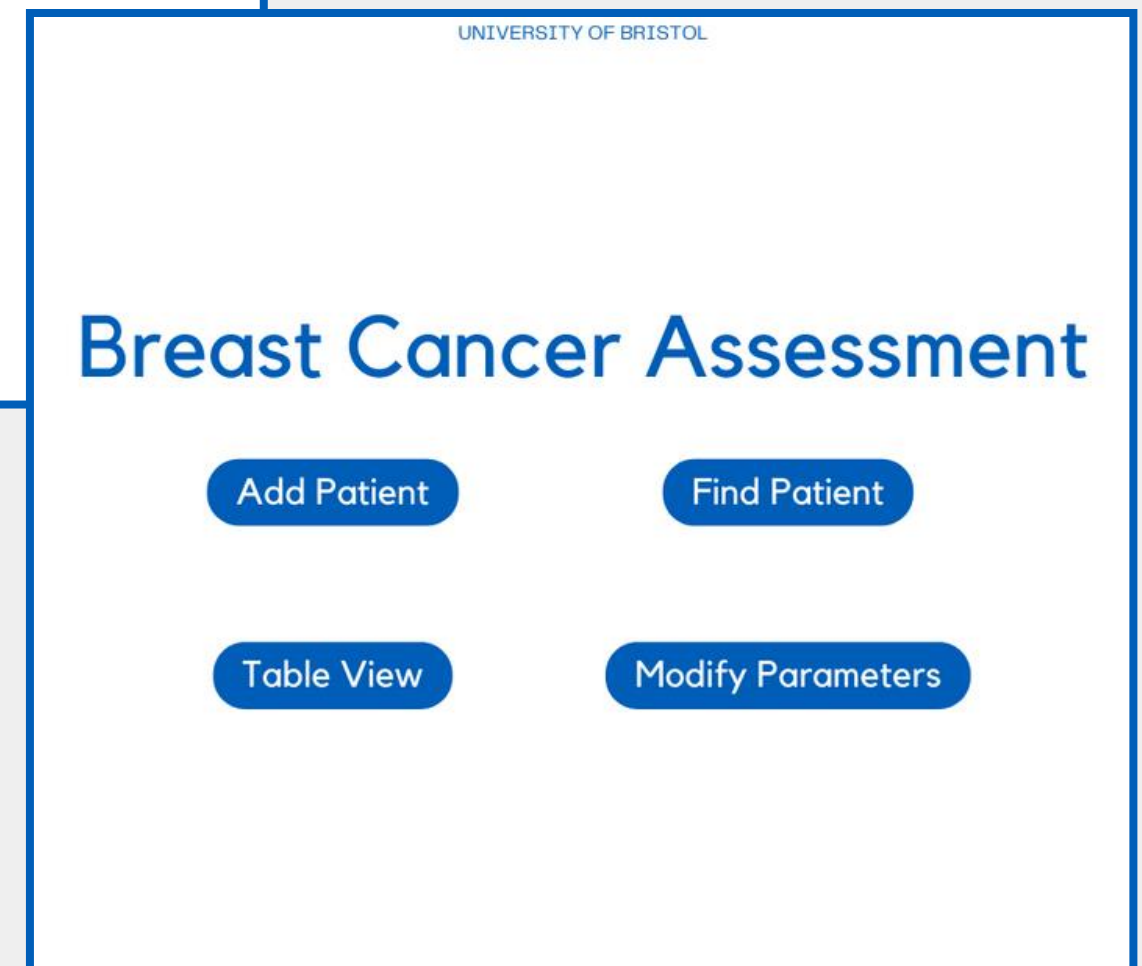
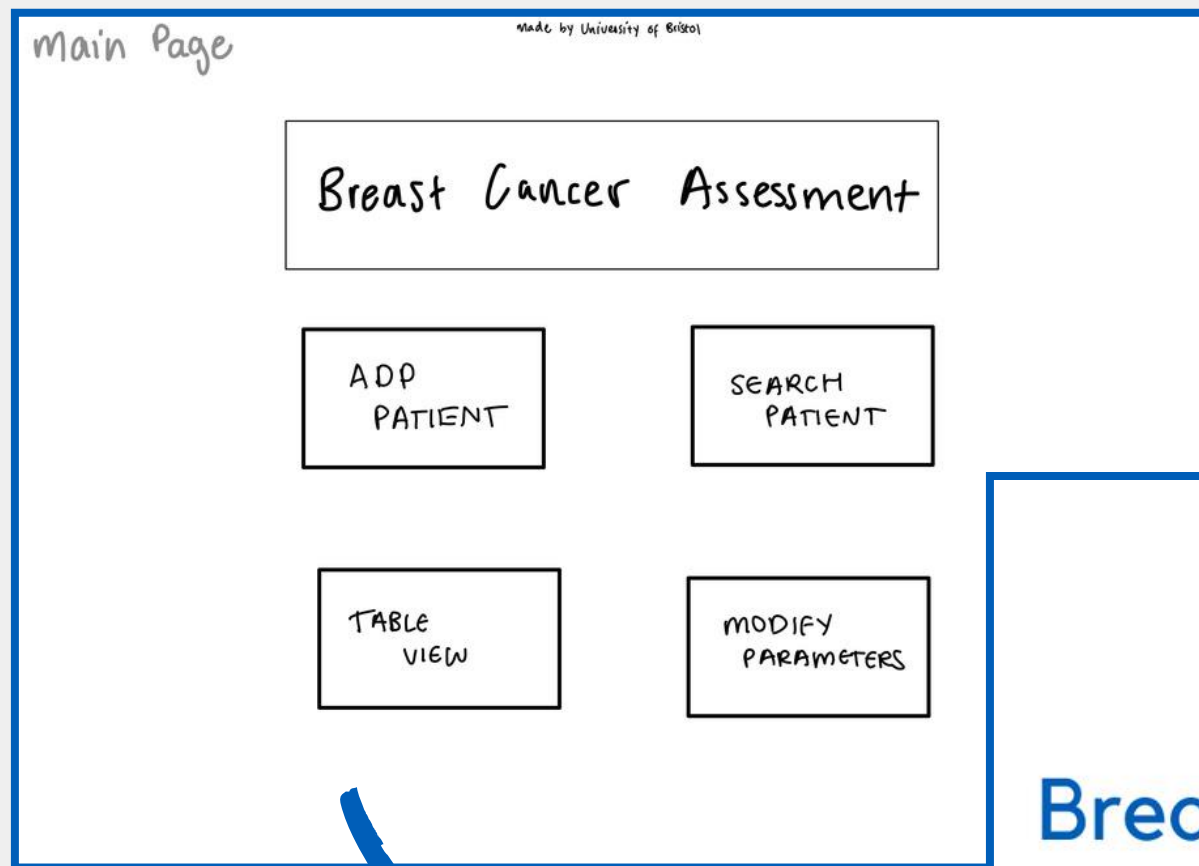
Timeline: TB1

1

From discussions with Nicky, we hand-drew a mock-design of the front end, and then made a UI prototype.

Priorities:

Simple & easy to use



Timeline: TB1

2

Began to create the foundation of the app and build some backend data structures such as patients and patient data (split into the 3/4 separate parts)



Timeline: TB1

3

Decided that we wanted to use Maps and JSON files for the data and algorithms respectively.

Opened a connection between back-end and front-end.

Added add patient functionality.



Timeline: TB2

1

Started **family history** and started theorising about what to do if we were told we could start to store **data**.



Timeline: TB2

2

Added data storage as well as implemented family history fully.

Then made sure all data is encrypted and **added Search Page and Table View.**



TB2 Testing Day: Feedback

- Clunky to use
- Too much information with entire form on one page. Hard to keep track of filled vs unfilled fields
- Too slow to use/not ergonomic. A lot of friction with input selection



TB2 Response to feedback

- Incorporated drop-boxes (previously had checkboxes)
- Incorporated tab structure, multiple tabs for different forms and type of tumour
- Typing selection for parameters with discrete options

The image displays three screenshots of a web form interface, illustrating changes made in response to feedback.

The top screenshot shows a section with three labels: "Previous SACT:", "Size:", and "Family History Risk:". To the right of "Previous SACT:" is a drop-down menu that is open, showing four options: "WITH_ANTHRA" (highlighted in blue), "WITHOUT_ANTHRA", "UNKNOWN", and "NO".

The middle screenshot shows a tabbed interface with three tabs: "Personal Details Form", "Pre Op Form", and "Post Op Form". The "Pre Op Form" tab is currently selected. Below the tabs is a "Back" button. To the right of the "Back" button, the text "Pre O" is partially visible.

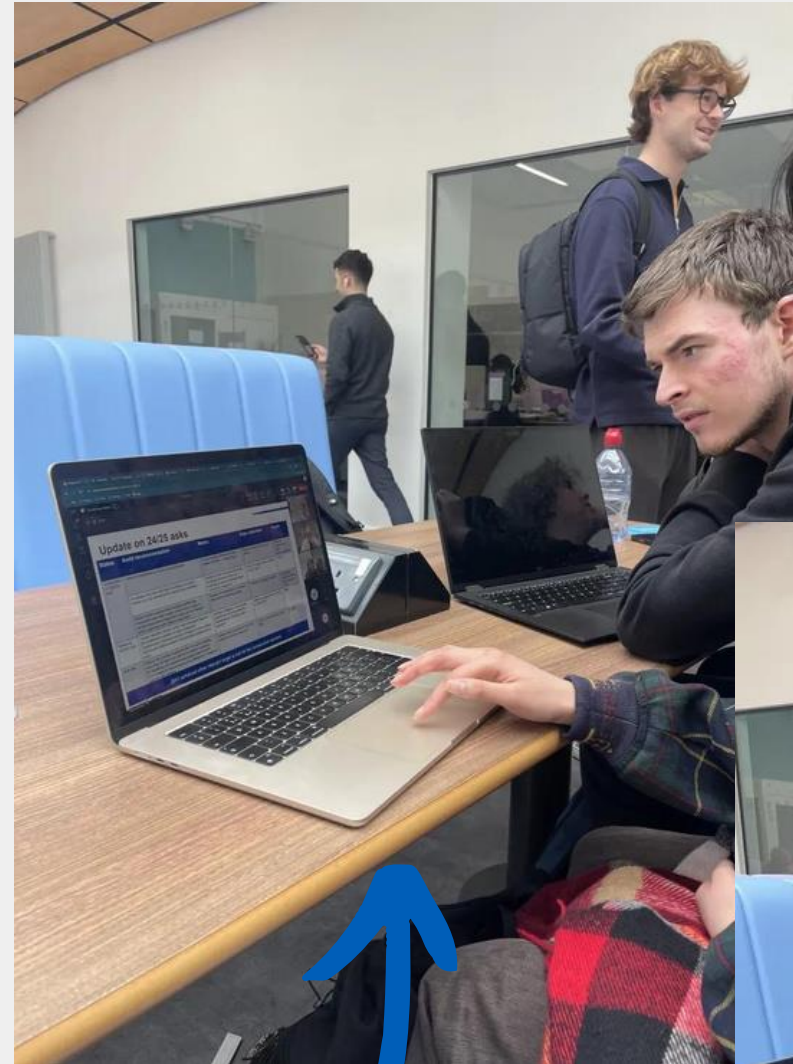
The bottom screenshot shows a section with two labels: "Family History Risk:" and "Histological Type of Breast Cancer:". To the right of "Family History Risk:" is a drop-down menu that is open, showing one option: "POPULATION".



TB2 Events

Attended the National Cancer Alliance Treatment VWG Conference on 7th March.

**Presenting
live online!**

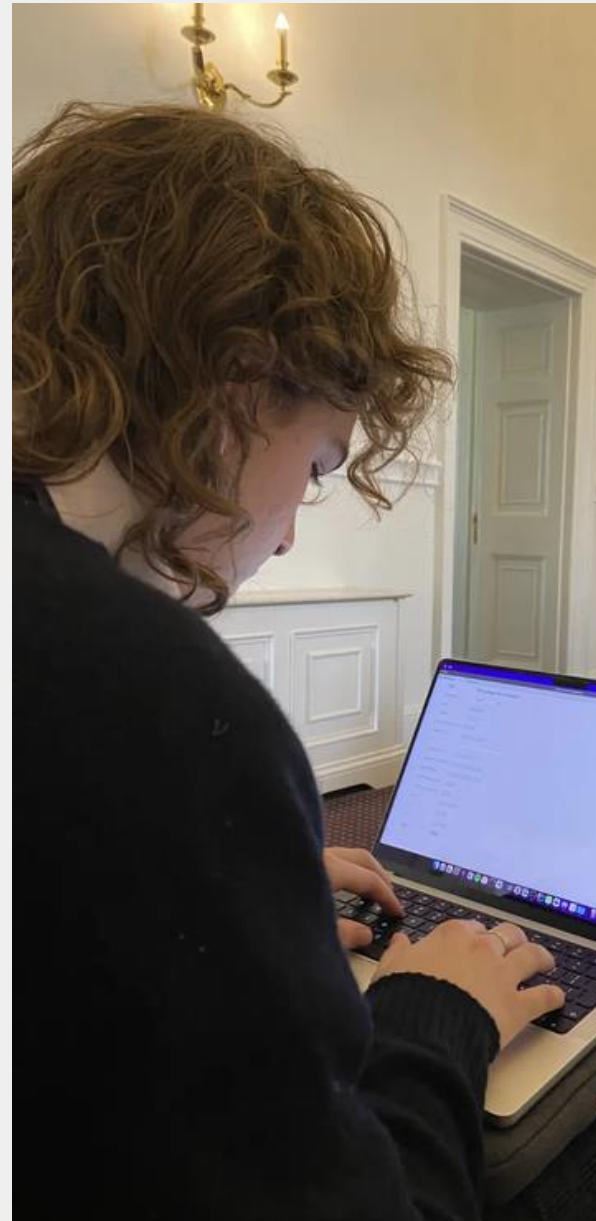


Demonstrated our project to a
audience of doctors and
researchers.
Received amazing support from
several hospitals and even the head
of the national cancer trust.



TB2 Events

Also attended the National
Cancer Treatment Alliance
Conference in Bath



TB2 Events

Soraya Conrad writing a research paper about using algorithmic programs for treatment eligibility + other future collaborations with university students



Set up and use of and low cost Algorithmic program to aid the breast MDT at RUH, Bath and Bristol Breast Unit, Southmead.

Soraya, Blaise, Josh, Maisie, Lamb, Elijah, Adib, Nikki, Charlie Comins , Rachel Ainsworth, Emma Keane, Abi Jenner, Rebecca Bowen Nicky Laurence

Introduction: The multidisciplinary team meeting has long been established to ensure cancer patients are offered best possible treatment pathways. The MDT is a costly meeting and with increased expertise, there is a need to streamline straightforward pathways. However, with increasing evidence for deescalation of treatments in certain circumstances , for example partial radiotherapy, and also an increased number of further therapies for high risk patients, the MDT is becoming increasingly complex with an increased number of inclusion and exclusion criteria that must be remembered and checked. This is time consuming and subject to human error.

Many of these criteria are similar and simple to record as parameters into an algorithmic program containing inclusion and exclusion criteria for each treatment option or investigation aligned with UK guidance for breast cancer treatments. With open source codes now freely available, a collaborative project with computer science university students was established. The need involves simple computer science coding while allowing second year students within the public sector to collaborate with the NHS public sector to provide tools of collaboration for the students, while providing a useful functional database.

Aim : To provide a functional database where patient data are accurately recorded into essential parameters. We set up a series of algorithms in order to give an output of investigations or treatments that patients would be eligible for without the need to check each criteria and to ensure minimal errors were made. The time to do this would be much decreased. These outputs are used with the raw data to aid the MDT discussion. The data would be stored for further investigations and treatments if required and for research and audit purposes.

Method:

By using these algorithmic programs, we have been able to ensure that all investigations and treatments are offered appropriately. That these decisions are aided in a timely manner and that few errors are made. In addition we have a long term database with extractable data across Bristol and the RUH and for any other hospital choosing to use this product. In addition, we have been able to train and work with a small group of computer science students to enable their learning to increase their employability following their degree. They will have an increased knowledge of medical IT and may choose this area in the future.

Future Goals

We meet with **AstraZeneca** on 6th June.

AZ are selling a new epigenetic drug for breast cancer treatment. Due to the high number of diagnostics needed to give it to a patient, the drug is not accessible to most busy hospitals.

Our program solves this.



Future Goals

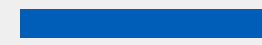
Nicky would like to continue working with us to roll the application out **nationwide.**

- 1 Implement “add parameters” functionality to allow doctors more autonomy over the application
- 2 Add “family history risk” calculator to accompany the Manchester Score functionality we have already implemented



Breast Cancer Assessment

Thank You!



Josh Chatten

Elijah Durston-Rand

Lamb Chen

Blaise Sheehan